The Local Housing-Market Effects of Act 22/60 Investor Purchases: Data and Empirical Design

July 2026

1 Research question

When a tax-incentivized migrant (an Act 22/60 "Individual Investor" decree holder) buys a residential property in Puerto Rico, what happens to the housing market immediately around that property — to prices, and to who transacts?

2 Data construction

2.1 Identifying investor properties

- **Decree holders.** Government lists of individual grantees: 3,181 (Act 22) and 2,672 (Act 60) names; 5,770 distinct first–last pairs pooled.
- **CRIM cadastral registry.** Name searches against the CRIM parcel layer (via its ArcGIS REST service) matched 26,673 unique parcels, each with full attributes, centroid coordinates, and the parcel's *most recent* sale (price, date, buyer, seller). CRIM stores only the latest transaction per parcel.
- **Karibe property registry.** Registry name-search results; cadastral numbers extracted from registral descriptions and resolved in CRIM provide a second, largely non-overlapping source.
- **Match confidence.** Name→parcel matching is noisy for common names. A name returning *exactly one* property is treated as high-confidence ("unique match"): 810 parcels via CRIM, 755 via Karibe (surname-validated, geocoded), union **1,306 parcels** (259 cross-validated in both). The event studies use this base.
- **Coverage.** High-confidence identification covers 22.5% of searched names (26.7% including a validated multi-match recovery tier), versus 49–61% declared homeownership in the Act 22 annual reports. The gap is dominated by LLC/trust purchases (invisible to name search), name-format failures, and registry availability; the identified sample skews toward single-property, personal-name buyers.

2.2 Events

An **event** is a dated investor purchase: unique-match parcels with a valid CRIM sale date, collapsing same-day, same-location purchases (condo units bought together) into one event, giving 1,256 events (1,200 in the decree era, $\geq$ 2012, used in estimation). Purchases cluster heavily: the median event has 25 other events within 1 km (158 are fully isolated), so contamination is measured per event and used in robustness.

2.3 Outcomes: nearby sales

For each event, every CRIM parcel within 1,000 m with a recorded sale: 3.47M sale$\times$event pairs over 328K distinct parcels, each carrying distance, sale price/date/parties, property characteristics (lot size, assessed structure value, condo-unit and vacant-land indicators), and 2020 census tract. *Rings:* 0–250 m (treated), 250–400 m (buffer, dropped), 400–1,000 m (control). *Filters:* nominal transfers ($<$ \$10k), implausible dates, and sales of investor-matched parcels are excluded. *Caveat:* CRIM keeps only each parcel's last sale, so earlier sales are censored by later resales; the within-window, near-versus-far design mitigates this, and Figure 5 tests it.

2.4 Placebo events

1,500 purchases by non-corporate *individual* buyers in the investor price band (\$285K–\$3.9M, the p25–p95 of investor purchases), located within 250 m of a real event site (where the sales pool has full coverage) but more than three years apart in time. Far rings are truncated at 750 m for both the placebo run and a matched real-event baseline.

2.5 Buyer/seller origin proxy

Party names are classified against the Census 2010 surnames file (`pcthispanic`), taking the maximum across name tokens (robust to CRIM's inconsistent name ordering; appropriate under the Puerto Rican two-surname convention). Names with `pcthispanic` ≤ 20 are "non-Hispanic-named" (mainland proxy), ≥ 70 "Hispanic-named" (local proxy); corporate, ambiguous, and unmatched names are left unclassified, and the raw score is retained for re-thresholding. *Validity:* 67% of investor purchases have non-Hispanic-named buyers versus 10% of surrounding market sales. *Caveat:* the proxy captures *name* origin — stateside Puerto Ricans and other Hispanic-named mainlanders classify as local, so mainland penetration and displacement effects are understated.

3 Empirical design

3.1 Panel construction

Because CRIM records one (the last) sale per parcel, sales are aggregated into an event$\times$ring$\times$period panel. Let e index events, $r \in \{\text{near}, \text{far}\}$ rings, and t calendar periods (years in the robustness suite). The cell outcome is the mean log price among the N_{ert} transactions in cell (e, r) in period t:

$$\bar{y}_{ert} \;=\; \frac{1}{N_{ert}} \sum_{j \in (e,r,t)} \ln P_j, \tag{1}$$

where P_j is the sale price of transaction j. Treatment is absorbing and turns on for the near cell in the event period t_e:

$$D_{ert} \;=\; \mathbf{1}\{r = \text{near}\} \times \mathbf{1}\{t \geq t_e\}, \tag{2}$$

so far cells are never treated and serve as clean controls. In the tract-fixed-effects specification, $\ln P_j$ is first residualized on census tract dummies at the sale level $(\ln P_j - \hat{\gamma}_{\tau(j)})$ and the residual is aggregated instead; results are unchanged.

3.2 Local projections difference-in-differences (LP-DiD)

Following Dube, Girardi, Jordà and Taylor (2023), the dynamic effect at each horizon h is estimated from a separate long-difference regression on the stacked cell panel. For post-treatment horizons $h = 0, 1, \ldots, H$:

$$\bar{y}_{er,\,t+h} - \bar{y}_{er,\,t-1} \;=\; \beta_h^{\text{LP-DiD}}\, \Delta D_{ert} \;+\; \delta_t^h \;+\; \varepsilon_{ert}^h, \tag{3}$$

and for pre-treatment horizons $h = 2, \ldots, Q$ (with $h = 1$ the normalization):

$$\bar{y}_{er,\,t-h} - \bar{y}_{er,\,t-1} \;=\; \beta_{-h}^{\text{LP-DiD}}\, \Delta D_{ert} \;+\; \delta_t^h \;+\; \varepsilon_{ert}^h, \tag{4}$$

where $\Delta D_{ert} = D_{ert} - D_{er,t-1}$ indicates that the near cell of event e enters treatment in period t, and δ_t^h are horizon-specific calendar-period effects.

Each horizon-h regression is estimated on the restricted "clean" sample

$$\Big\{ (e, r, t) : \quad \underbrace{\Delta D_{ert} = 1}_{\text{newly treated near cells}} \quad \text{or} \quad \underbrace{D_{er,s} = 0 \;\; \forall s}_{\text{never-treated far cells}} \Big\}, \tag{5}$$

which is the `nevertreated` option: the counterfactual change $\bar{y}_{er,t+h} - \bar{y}_{er,t-1}$ for a newly treated near cell is identified from far cells over the same calendar window, and no already-treated observation is ever used as a control (the negative-weighting problem of two-way fixed effects under staggered timing cannot arise). $\beta_h^{\text{LP-DiD}}$ is a variance-weighted average of event-specific effects; standard errors are clustered by cell. The identifying assumption is within-micro-neighborhood parallel trends: absent the purchase, prices of homes transacting 0–250 m from the site would have evolved like those 400–1,000 m away, conditional on calendar-time effects — shocks common to the micro-area (Hurricane María, the 2020–21 in-migration wave) difference out by construction, and equations (4) provide the testable pre-trend restrictions $\beta_{-h} = 0$.

Two scope conditions follow from the ring structure. First, the near–far contrast differences out any effect *common to the full 1-km neighborhood*: the design identifies the hyper-local increment around a purchase site, not the neighborhood-level effect of aggregate investor inflows, which requires the complementary tract-level staggered design using the tract×month price panel. Second, because purchases cluster at the enclave scale, other investor purchases in dense areas fall disproportionately inside the near ring itself; the estimated contrast there measures the cumulative local dose of investor activity rather than the effect of a single purchase (see Figure 4).

In the annual robustness suite $Q = 4$ and $H = 3$; the baseline is also estimated at quarterly ($Q = H = 8$) and half-yearly resolutions and with the Callaway–Sant'Anna (2021) group-time estimator $ATT(g,t) = \mathbb{E}[\bar{y}_t - \bar{y}_{g-1} \mid G_g = 1] - \mathbb{E}[\bar{y}_t - \bar{y}_{g-1} \mid \text{never treated}]$, aggregated to event time; results are similar across estimators and resolutions, and identical with and without tract fixed effects.

4 The eight figures

All figures plot LP-DiD event-study coefficients $\{\beta_{-Q}, \ldots, \beta_H\}$ from equations (3)–(4) at annual resolution (pre-window 4, post-window 3), with 95% confidence intervals, cells clustered. Period -1 is the omitted base; the dashed vertical line marks the treatment boundary.

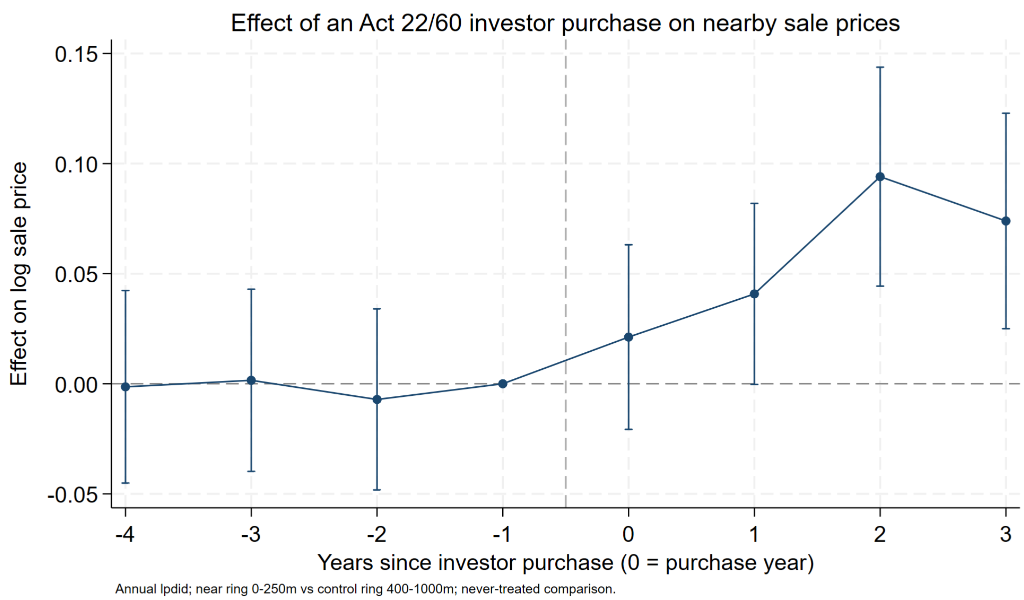

Figure 1: **Baseline event study.** Near-ring (0–250 m) versus far-ring (400–1,000 m) effects on log sale price by year relative to the investor purchase. The headline price bump; flat pre-period coefficients are the design's core credibility check, and the post-period path shows how quickly the premium appears and whether it persists.

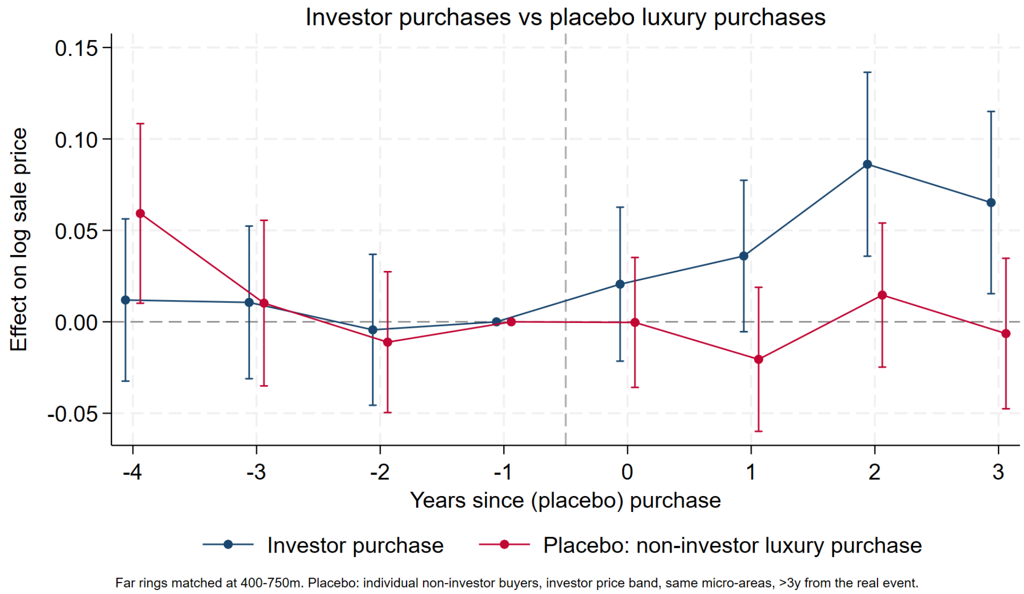

Figure 2: **Investor vs. placebo purchases.** The real event study overlaid with placebo events (non-investor individual buyers in the investor price band, same micro-areas, >3 years from the real event; far rings matched at 400–750 m). A flat placebo says nearby prices respond to *the investor*, not to any high-value transaction; because unmatched investors contaminate the placebo toward finding an effect, a null placebo is conservative evidence of investor specificity.

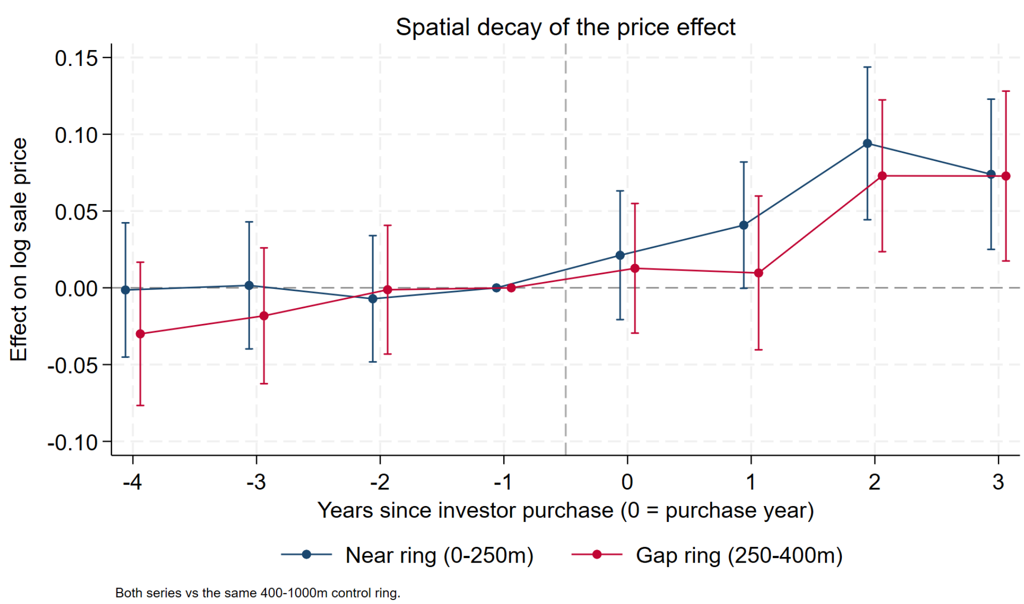

Figure 3: **Spatial gradient.** Near-ring (0–250 m) and gap-ring (250–400 m) series, each against the same far control. A genuine hyper-local spillover should decay with distance (near > gap > 0); near ≈ gap would instead indicate a neighborhood-wide shock partially shared by the control ring.

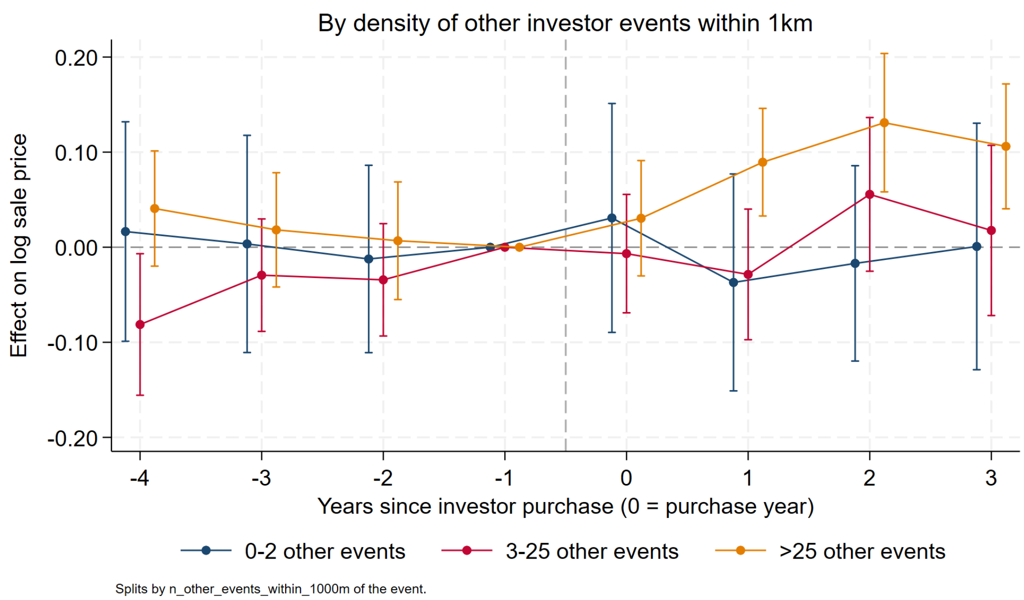

Figure 4: **Estimated contrast by local event density.** Separate event studies for events with 0–2, 3–25, and >25 other investor events within 1 km. Because purchases cluster at the enclave scale (condo buildings, resort communities), other events in dense areas fall disproportionately *inside* the near ring, so the near–far contrast there reflects cumulative local investor exposure rather than a single purchase. The estimated effect is increasing in density and near zero for isolated events: single-purchase channels (comps entry, one renovation, signaling) appear weak on their own, and the price response is a phenomenon of investor *concentration*. Per-event causal attribution is accordingly not warranted in the dense group, where the coefficient measures the differential appreciation of investor-saturated micro-pockets relative to their 400–1,000 m surroundings — and where selection of investor waves into already-appreciating pockets is the leading alternative reading (see pre-period coefficients by group).

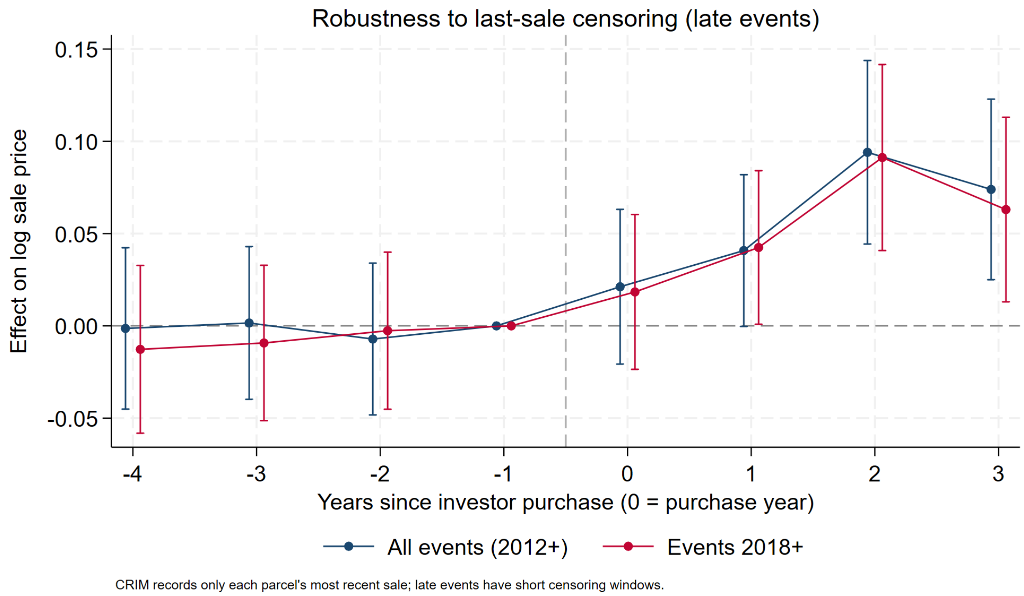

Figure 5: **Censoring robustness.** Baseline overlaid with events from 2018 onward, for which little time exists for post-period sales to be overwritten by later resales. Agreement with the full sample indicates that CRIM's last-sale-only censoring is not manufacturing the bump.

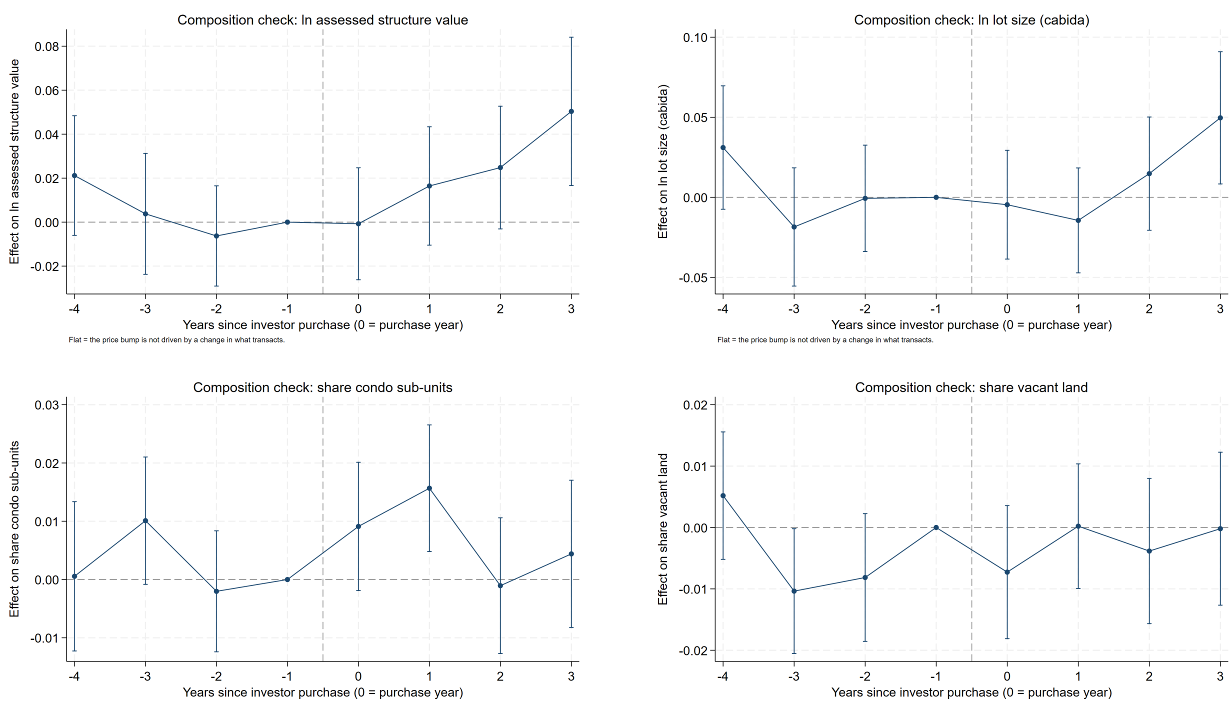

Figure 6: **Composition checks.** The identical design with property characteristics as outcomes: log assessed structure value (top left), log lot size (top right), condo-unit share (bottom left), vacant-land share (bottom right). Flat lines imply the price bump is appreciation on a stable transaction mix; movements would reveal the stock-upgrading margin of gentrification, changing the interpretation of Figure 1.

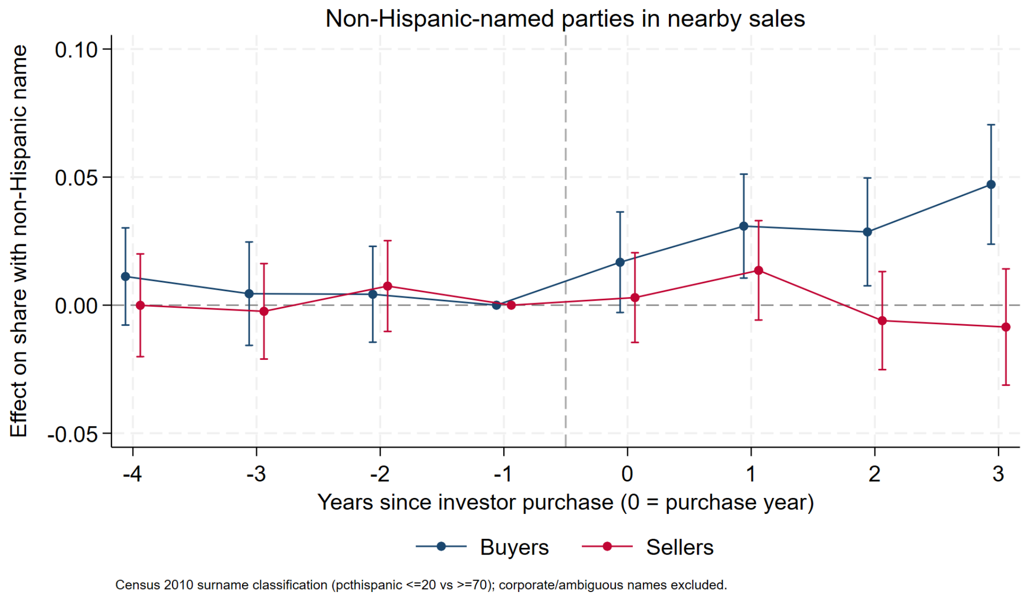

Figure 7: **Who transacts: buyers vs. sellers.** Effects on the share of nearby sales with non-Hispanic-named buyers and, separately, sellers (Census 2010 surname classification; corporate and ambiguous names excluded). Buyer share rising faster than seller share means the area is net-importing mainland owners after an investor arrives — the composition shift price indices cannot see.

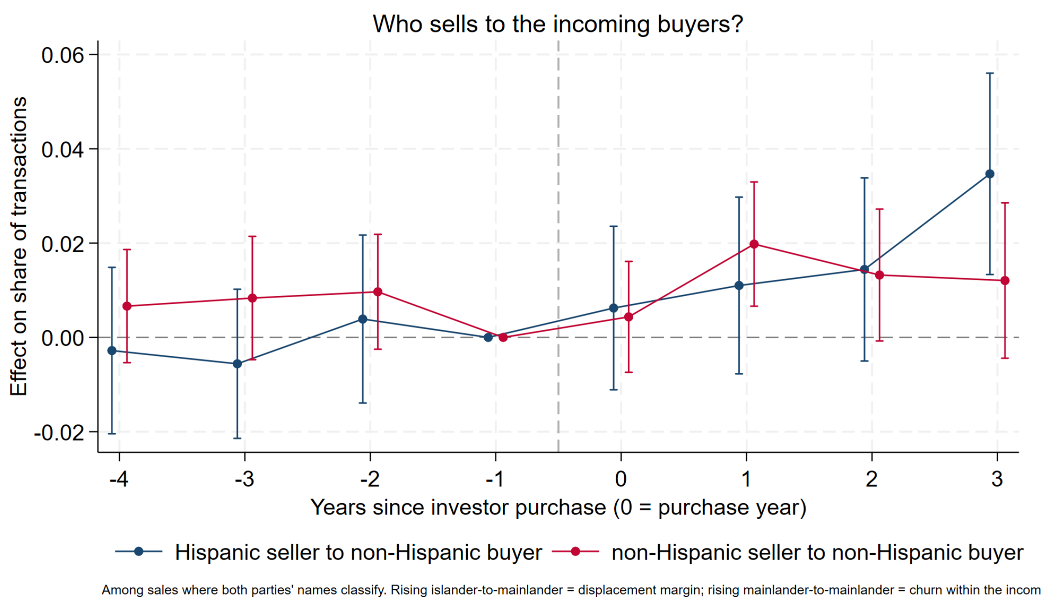

Figure 8: **Transition types.** Effects on the shares of transactions that are Hispanic-seller→non-Hispanic-buyer versus non-Hispanic-seller→non-Hispanic-buyer (among sales where both parties classify). Rising islander→mainlander transitions indicate conversion of locally held housing to incomers (the displacement margin); rising mainlander→mainlander indicates churn within an already-converted segment — together distinguishing “investors lead conversion” from “investors cluster where conversion already occurred.”

5 Standing caveats

1. **Last-sale censoring** (CRIM): pre-period sales are survivors that never resold; addressed by design and Figure 5.
2. **Site selection**: investors choose appreciating micro-areas; estimates are local spillovers *conditional on* location choice, not "the effect of the Act."
3. **Match selection**: identified investors skew toward single-property, personal-name buyers; entity (LLC/trust) purchases are missing.
4. **Name proxy**: understates mainland presence; corporate and ambiguous names are excluded from shares.
5. **Assessed values** are on CRIM's 1957 basis — valid as cross-sectional controls and composition outcomes, never as market values.